

L9: [Chapter_3. Data Service Integration with **REST Java Architecture: J4DI**: Integration and Analytical with SparkSQL]

Data Services Architecture (DSA) Study Case: source directory structure

- Support Java Projects (Maven) directory
 - fdbo_study_case/dsa_study_case
- DOC Access Model projects (archived)
 - **DSA-SQL-JDBCService.zip**
 - **DSA-SQL-JPADBService.zip**
 - **DSA-DOC-XLSService.zip**
 - **DSA-DOC-XMLService.zip**
 - (alternative) **DSA-NoSQL-MongoDBService.zip**
 - (alternative) **DSA-NoSQL-Neo4JService.zip**
- SparkSQL Standalone Server as SpringBoot Service
 - **DSA-SparkSQL-Service**

PREPARE Data Source SparkSQL Remote Views

- Start **Source Data Sservices** (microservice SpringBoot architecture)
 - DSA-SQL-JDBCService
 - SpringBootSQLJDBCService class
 - DSA-SQL-JPADBService
 - SpringBootSQLJPADDataService class
 - DSA-DOC-XLSService
 - SpringBootDOCXLSService class
 - DSA-DOC-XMLService
 - SpringBootDOCXMLService
- Check/Reload **SparkSQL** views for REST Endpoints (same procedure as in IIS_Lab_6)
 - Run **SparkSQL** server *standalone*
 - Open Java Project Module [DSA-SparkSQL-Service]
 - Open SBT application class *SprinBootSparkSQLStarter* from
 - package org.spark.service
 - Check *SprinBootSparkSQLStarter* Run Configuration
 - **RUN** @SpringBootApplication SprinBootSparkSQLStarter
 - **Connect to SparkSQL server with DBEaver Spark Connection**(use the connection created in IIS_Lab_6)
 - File → Open: src/main/resources/scripts from Data Source Access projects

- **DSA-SQL-JDBCService**
 - **DS1_PostgreSQL_SparkSQL_Views.sql**
 - **DSA-SQL-JPADBService**
 - **DS2_ORACLE_SparkSQL_Views.sql**
 - **DSA-DOC-XLSService**
 - **DS4_XLSX_SparkSQL_Views.sql**
 - **DSA-DOC-XMLService**
 - **DS3_XML_SparkSQL_Views.sql**
 - Run the SQL Scripts
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CASE STUDY 1: Case Study: SparkSQL Multidimensional ROLAP Database

- Create Multidimensional Integration and analytical Views on SparkSQL schema
 - DBeaver (Connect to SparkSQL - use the connection created in IIS_Lab_6)
 - From **DSA-SparkSQL-Service project**, into *src/main/resources/scripts*:
 - **SparkSQL_OLAP_Multidimensional_Analytical.sql**
 - Create
 - *Consolidation Views*
 - *Dimensional Views*
 - **OLAP_DIM_CUSTS_CITIES_DEPTS**
 - **OLAP_DIM_DATA_CALENDAR**
 - **OLAP_DIM_CUST_CTG_TO**
 - **OLAP_DIM_CUST_CTG_EMP**
 - **PRODUCTS_VIEW** (from *DS2_ORACLE_SparkSQL_Views.sql*)
 - **PERIODS_VIEW** (from *DS4_XLSX_SparkSQL_Views.sql*)
 - *Fact Views*
 - **OLAP_FACTS_SALES_AMOUNT**
 - *Analytical Views*
 - *Aggregation (CUBE, ROLLUP) views*
 - **OLAP_VIEW_SALES_DEP_CIT_CUST**
 - **OLAP_VIEW_SALES_CALENDAR**
 - **OLAP_VIEW_SALES_CTG_CUST_TO**
 - **OLAP_VIEW_SALES_CTG_CUST_EMP**
 - **OLAP_VIEW_SALES_CTG_PROD**
 - **OLAP_VIEW_SALES_CTG_PROD_CITIES**
- Test Web View REST endpoints on SparkSQL schema (**Analytical Web Model**)
 - Auto-activated *REST Views* (by *org.spark.service.rest*. **SparkSQLRESTService** from SparkSQL-SBT distribution)
 - *Analytical Views*, e.g.

- OLAP_VIEW_SALES_DEP_CIT_CUST
- OLAP_VIEW_SALES_CALENDAR
- OLAP_VIEW_SALES_CTG_CUST_EMP
- OLAP_VIEW_SALES_CTG_CUST_TO
- OLAP_VIEW_SALES_CTG_PROD_CITIES

- **Endpoint Views**

http://localhost:9990/DSA-SparkSQL-Service/rest/view/OLAP_VIEW_SALES_DEP_CIT_CUST

http://localhost:9990/DSA-SparkSQL-Service/rest/view/OLAP_VIEW_SALES_CTG_PROD_CITIES

http://localhost:9990/DSA-SparkSQL-Service/rest/view/OLAP_VIEW_SALES_CALENDAR

http://localhost:9990/DSA-SparkSQL-Service/rest/view/OLAP_VIEW_SALES_CTG_CUST_EMP

http://localhost:9990/DSA-SparkSQL-Service/rest/view/OLAP_VIEW_SALES_CTG_CUST_TO